

Building Your Own Storage Area Network and Network Attached Storage

No matter how much storage is available, there's always a need for more. Storage solutions are a must in the modern world, especially with the explosion of data. This article introduces readers to a few open source storage solutions, and walks them through the process of building their own SAN and NAS solutions.



Generally, storage solutions can be grouped into the following four categories: SOHO NAS systems, cloud based/object solutions, Microsoft storage server solutions, enterprise NAS (networked attached storage) and storage area network (SAN) solutions.

Enterprise NAS and SAN solutions are generally closed systems offered by traditional vendors like EMC and NetApp and come with a very large price tag; hence, many businesses are looking at open source solutions to meet their needs. So here is a collection of links and brief descriptions of the open source storage solutions currently available.

Open source of course means it's free to use and modify; however, some projects do have commercially supported versions as well, for enterprise customers who require these.

Factors for choosing storage solutions

When you are in need of storage solutions to manage your own infrastructure in your private data centre, there are many offerings available, but selecting the right one depends upon your requirements.

Factors involved in selecting a storage type are:

- Budget

- Type of data to be stored
- Scaling concerns
- Usage pattern

In this article, we will discuss two different methods (NAS and SAN), which define the structure of the storage. It is important to choose the right one that suits your use case and type of data.

Network attached storage

A network attached source (NAS) is a device or server attached to the TCP/IP network, and the data stored in this device can be shared. When a file read/write is requested in the form of a Network File System (NFS) or Common Internet File System (CIFS), it is sent to the NAS attached to a TCP/IP server, which, on receiving the NFS or CIFS request, converts the request into a local storage IO command set.

Features of NAS: NAS uses file level data access. The main I/O protocol of NAS is NFS or CIFS. NAS may appear like a 'shared folder' but it is connected by means of the Ethernet, which in some cases can worsen the execution if the system is being used for different purposes (which is typically